

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

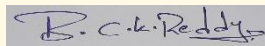
ASSESSMENT TOOL 1 – Analytical Problem Solving

Course Code:	ITA0512	Course Title:	Computer Vision
CO Assessed:	CO2 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)		
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Section / Batch:	CSE – Ai / 2024	Date:	08/08/2026

Code of Conduct

I B. Charan Kumar Reddy (192472195) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Criteria	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	Problem Understanding	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	Method Selection	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	Solution Process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	Accuracy of Calculation	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	Interpretation of Results	

Name: B. Charan Kumar Reddy

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Course Code: ITA 0512

Course Name : Computer Vision

Answers

Question 4: Spatial Filtering (Smoothing)

4(a). Explain the purpose of spatial filtering for smoothing in digital image processing.

Definition

Spatial filtering is an image processing technique in which a filter (kernel or mask) moves across an image and modifies the value of each pixel based on its neighboring pixels. Smoothing filters are designed to reduce noise and unwanted intensity variations while preserving the overall image structure.

Purpose of Smoothing

The primary purpose of smoothing is to improve image quality by reducing random noise and small intensity fluctuations. It helps in preparing images for further processing such as segmentation, feature extraction, object recognition, and edge detection.

Working Principle

A smoothing filter replaces the center pixel with the average or weighted average of its neighboring pixels. This process minimizes sudden intensity changes and produces a smoother image.

Common Smoothing Filters

1. Mean (Average) Filter
2. Gaussian Filter
3. Median Filter
4. Bilateral Filter

Advantages

- Removes random noise effectively.
- Reduces unwanted intensity variations.
- Produces smoother images.
- Improves preprocessing for image analysis.
- Enhances overall image quality.

Disadvantages

- Fine image details may be lost.
- Image edges may become blurred.
- Excessive smoothing reduces sharpness.

Applications

- Medical image preprocessing
- Satellite image enhancement
- Face recognition systems
- Self-driving vehicles
- CCTV surveillance systems

Conclusion

Spatial smoothing is an essential preprocessing technique in digital image processing because it removes noise, improves image quality, and prepares images for accurate feature extraction and analysis.

4(b). Apply a 3×3 Averaging Filter on the Given Image Matrix.

Given Image Matrix

$$\begin{bmatrix} 10 & 20 & 30 \\ 20 & 30 & 40 \\ 30 & 40 & 50 \end{bmatrix}$$

Step 1: Averaging Filter

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Step 2: Calculate the Sum

$$10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50$$
$$= 270$$

Step 3: Find the Average

Average

$$= 270 / 9$$

$$= 30$$

Result

The new center pixel value after applying the averaging filter is

30

Interpretation

The averaging filter replaces the center pixel with the average value of its neighboring pixels. This reduces random noise and smooths the image while slightly reducing edge sharpness.

Advantages of Averaging Filter

- Simple to implement.
- Reduces Gaussian noise.
- Improves visual quality.
- Fast computational performance.

Limitations

- Blurs image edges.
- Removes fine image details.
- Not effective for salt-and-pepper noise.

Final Answer

Output Pixel = 30

Question 5: Spatial Filtering (Sharpening)

5(a). Explain how spatial filtering is used for sharpening an image.

Definition

Spatial filtering for sharpening is an image enhancement technique that highlights fine details, edges, and object boundaries by emphasizing high-frequency components in an image. It increases the contrast between neighboring pixels, making the image appear clearer and sharper.

Purpose of Sharpening

The main purpose of sharpening is to improve the visibility of edges and fine details that may become blurred due to image acquisition, smoothing, or compression.

Working Principle

Sharpening filters calculate the intensity difference between a pixel and its neighboring pixels. Pixels with large intensity differences are enhanced, making edges more prominent. The most commonly used sharpening filters are based on the **Laplacian operator** and **High-Pass filters**.

Common Sharpening Filters

- Laplacian Filter
- High-Pass Filter
- Unsharp Masking
- High-Boost Filtering

Advantages

- Enhances image edges.
- Improves fine details.
- Increases image sharpness.
- Helps in feature extraction.
- Improves object recognition.

Disadvantages

- Amplifies image noise.
- May create false edges.
- Excessive sharpening reduces image quality.

Applications

- Medical imaging (tumor detection)
- Satellite image analysis
- Fingerprint recognition
- Self-driving vehicle vision
- Industrial quality inspection

Conclusion

Spatial sharpening improves the visual quality of an image by emphasizing edges and fine details. It is widely used in digital image processing for accurate analysis and feature extraction.

5(b). Apply the Laplacian Mask to the Given Image Matrix.

Given Image Matrix

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

Laplacian Mask

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Step 1: Multiply the Center Pixel

$$4 \times 20 = 80$$

Step 2: Multiply the Neighboring Pixels

Top Pixel

$$-1 \times 10 = -10$$

Bottom Pixel

$$-1 \times 10 = -10$$

Left Pixel

$$-1 \times 10 = -10$$

Right Pixel

$$-1 \times 10 = -10$$

Step 3: Add the Values

$$80 - 10 - 10 - 10 - 10 = 40$$

Output Pixel

Interpretation

The Laplacian filter detects rapid intensity changes around the center pixel. Since the center pixel has a much higher value than its neighboring pixels, the output is positive (**40**), indicating the presence of an edge. This enhances image sharpness and makes object boundaries more visible.

Advantages of Laplacian Filter

- Detects edges accurately.
- Improves image sharpness.
- Simple to implement.
- Useful for feature extraction.

Limitations

- Sensitive to image noise.
- Produces double edges in some cases.
- Often combined with smoothing filters for better performance.

Final Answer

Output Pixel = 40

Question 6: Frequency Domain Filtering

6(a). Compare Low-Pass Filter and High-Pass Filter in the Frequency Domain.

Definition

Frequency domain filtering is an image enhancement technique in which an image is transformed from the spatial domain to the frequency domain using the **Fourier Transform (FT)**. Filtering is then applied to modify the frequency components before converting the image back to the spatial domain using the **Inverse Fourier Transform (IFT)**.

Low-Pass Filter (LPF)

A Low-Pass Filter allows **low-frequency components** to pass while blocking high-frequency components. Since low frequencies represent smooth intensity variations, LPF produces a smoother image.

Characteristics

- Removes high-frequency noise.
- Produces smooth images.
- Reduces sharp edges.
- Preserves overall image structure.
- Used for image smoothing and noise reduction.

Applications

- Medical image denoising
- Satellite image preprocessing
- Noise removal before segmentation
- Image restoration

High-Pass Filter (HPF)

A High-Pass Filter allows **high-frequency components** to pass while blocking low-frequency components. High frequencies represent edges and fine details.

Characteristics

- Enhances edges.
- Increases image sharpness.
- Highlights fine details.
- Removes smooth background.
- Used for edge detection and image sharpening.

Applications

- Face recognition
- Object detection
- Fingerprint enhancement
- Industrial inspection
- Self-driving vehicle vision

Comparison Table

Low-Pass Filter	High-Pass Filter
Passes low frequencies	Passes high frequencies
Removes noise	Enhances edges
Produces smooth images	Produces sharp images
Reduces image details	Preserves fine details
Used for smoothing	Used for sharpening
Blurs image slightly	Increases image contrast

Advantages of Frequency Domain Filtering

- Fast processing for large images.
- Better frequency analysis.
- Effective noise removal.
- Accurate edge enhancement.
- Widely used in medical and satellite imaging.

Conclusion

Low-Pass Filters are mainly used for **noise reduction and image smoothing**, whereas High-Pass Filters are used for **edge enhancement and image sharpening**. The selection of the filter depends on the image processing application.

6(b). Apply a Low-Pass Filter to the Given Frequency Components.

Given Frequency Components

$$F = [100, 80, 40, 20]$$

Step 1: Identify Low-Frequency Components

The first two values represent the **low-frequency components**.

$$100, 80$$

The remaining values represent the **high-frequency components**.

$$40, 20$$

Step 2: Apply Low-Pass Filter

The Low-Pass Filter retains only the low-frequency components and removes the high-frequency components by replacing them with **0**.

$$F = [100, 80, 0, 0]$$

Step 3: Filtered Output

$$F = [100, 80, 0, 0]$$

Interpretation

The Low-Pass Filter removes the high-frequency components responsible for edges and rapid intensity changes. As a result, the filtered image becomes smoother with reduced noise, but some fine details and edge information are lost.

Advantages

- Reduces random noise.
- Produces smooth images.
- Improves image quality.
- Suitable for preprocessing.
- Simple and efficient.

Limitations

- Removes fine details.
- Blurs image edges.
- Not suitable when edge preservation is required.

Applications

- Medical image enhancement.
- Satellite image processing.
- Image restoration.
- Noise reduction before segmentation.
- Computer vision preprocessing.

Final Answer

Filtered Frequency Components

[100,80,0,0]